

Frage 1

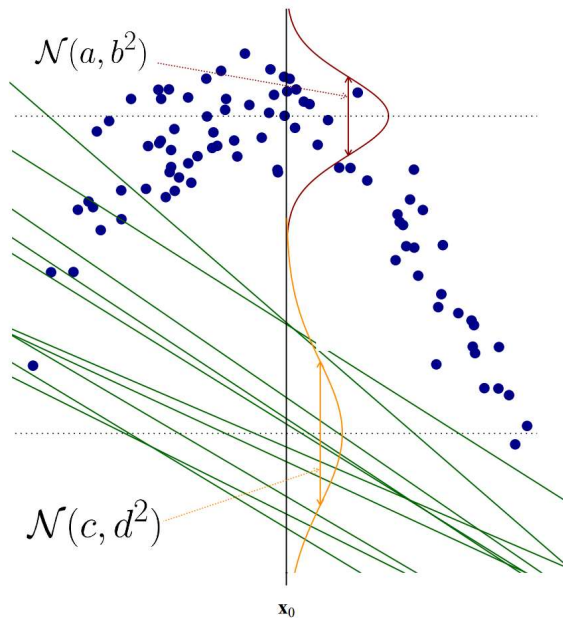
Richtig

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Bias variance decomposition for quadratic loss

Consider the following data points and model distributions, similar to what you saw in the lecture.

The vertical axis is the y-axis (corresponding to the labels). The horizontal axis is the x-axis of a single feature. The green lines are different trained models. The blue dots are sampled data points. $\mathcal{N}(\mu, \sigma^2)$ denotes the 1-d normal distribution with mean μ and variance σ^2 . The brown and orange lines depict the underlying distributions of labels (brown) and model predictions (orange), respectively, at a given input x_0 .



We also use the usual notation from the lecture, i.e.:

Let Z_l be a sample set of l elements and $g(\mathbf{x}_0; \mathbf{w}(Z_l))$ denote a trained model evaluated at $\mathbf{x}_0$, where the model parameters $\mathbf{w}$ are dependent on the respective training set Z_l .

The variables from the plot are:

$$a = 4$$

$$b = 3$$

$$c = -1$$

$$d = 5$$

Compute the values of the following terms:

- What is the unavoidable error at x_0 ? Answer: ✓
- What is the bias $E_{y|x_0}(y) - E_{Z_l}(g(\mathbf{x}_0; \mathbf{w}(Z_l)))$ at x_0 ? Answer: ✓
- What is the model variance at x_0 ? Answer: ✓
- What is the expected prediction error at x_0 ? Answer: ✓

Hint: the solutions are integers.

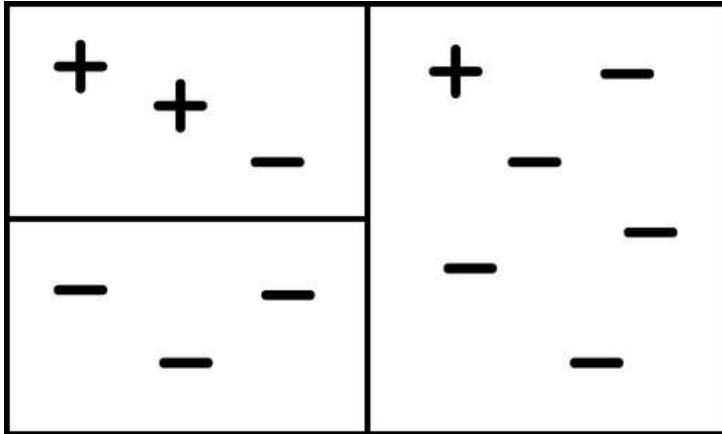
Frage 2

Falsch

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Gini impurity

Consider fitting a decision tree on binary data with classes + and -. Assume an intermediate iteration has resulted in the depicted partition.



What is the Gini purity gain of the next optimal partition?

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Do not give the answer in %. If your Moodle system language is set to English, use dot "." If your Moodle system language is set to German, use comma ",".

Antwort: 0,13



Frage 3

Teilweise richtig

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Sequence models

Which statements are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. One of the main limitations of transformers is that their computational demand scales exponentially in the sequence length.
- ☐ b. We denote with $\mathbf{R}$ the recurrent weight matrix at time t , with f the activation function, and with $\mathbf{s}(t-1)$ the preactivation at time $t-1$. For stable gradients, we need $\|\mathbf{R} \text{diag}(f(\mathbf{s}(t-1)))\| \approx 1$.
- ☒ c. RNNs are Turing complete. ✓
- ☒ d. In an RNN, the current hidden state depends on the current input and the previous output. ✗
- ☐ e. The attention mechanism is invariant to the simultaneous permutation of keys and values.
- ☒ f. In the attention mechanism, queries and keys have the same feature dimension but different lengths. ✓
- ☐ g. Classifying the sentiment of a text is a good example of a many-to-many task in sequence learning.
- ☒ h. LSTMs use an integrator to store information and gates for remembering, communicating, and (potentially) forgetting. ✓

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Richtig

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Area under curve

Consider the following classification results of a binary classifier tested on 8 samples:

$\bar{g}(x)$	y
0.1	1
-0.2	-1
2.0	1
-0.8	-1
-1.2	-1
1.5	1
-0.3	1
0.9	-1

Compute the Area Under Curve (AUC) of the ROC curve.

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Do not give the answer in %. If your Moodle system language is set to English, use dot "." If your Moodle system language is set to German, use comma ",".

Antwort: 

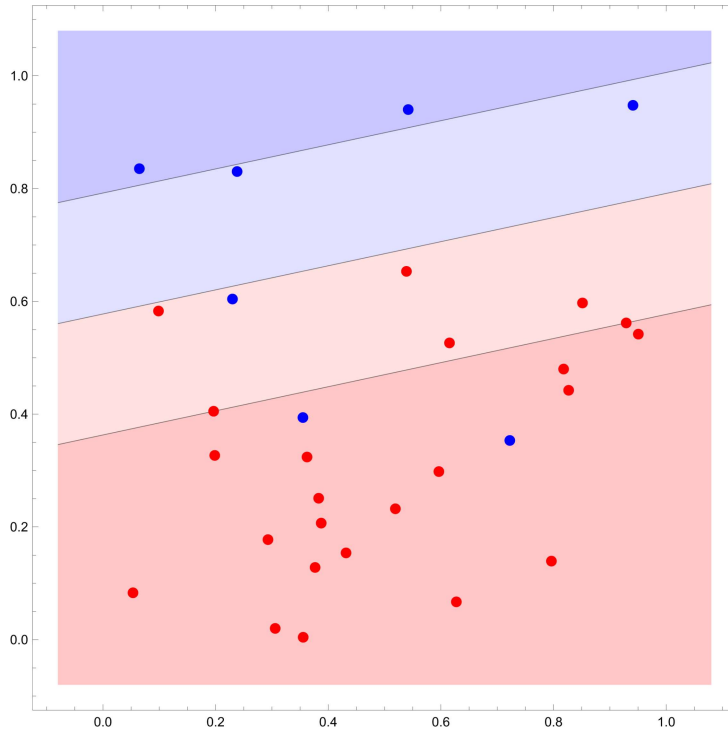
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Richtig

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Support Vector Machines

Consider a C-SVM classifier trained with a linear kernel on a simple 2D data set (red dots = positive samples, blue dots = negative samples) as depicted below. Blue areas are classified as negative (light blue = inside margin) and red areas are classified as positive (light red = inside margin).



How many data points are support vectors?

Number of blue support vectors: ✓

Number of red support vectors: ✓

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Richtig

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Boosting

For a given data set, an algorithm has found a sequence of weak binary classifiers g_n with $n = 1, \dots, 6$.

For a data point $\mathbf{x}$, the outputs of the individual weak classifiers and their respective weight factors are:

$$g_1(\mathbf{x}) = -1, \quad \alpha_1 = 2.5$$

$$g_2(\mathbf{x}) = +1, \quad \alpha_2 = 0.5$$

$$g_3(\mathbf{x}) = -1, \quad \alpha_3 = 2.5$$

$$g_4(\mathbf{x}) = +1, \quad \alpha_4 = 1.5$$

$$g_5(\mathbf{x}) = +1, \quad \alpha_5 = 2.0$$

$$g_6(\mathbf{x}) = -1, \quad \alpha_6 = 0.5$$

What is the output of the binary strong (boosted) classifier?

Antwort: ✓

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Convolution

Consider the input matrix

$$\mathbf{x} = \begin{pmatrix} 2 & 1 & 2 & 0 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 2 & 0 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 3 & 2 & 1 \end{pmatrix}$$

and the filter

$$\mathbf{W} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

- Assuming stride $S = 1$ and no padding, compute the top right entry $s_{1,5}$ of $\mathbf{s} = \mathbf{W} * \mathbf{x}$.

$$s_{1,5} = \boxed{-2} \quad \checkmark$$

- Assuming stride $S = 2$ and symmetric padding of size $P = 2$, the dimension of the output matrix is $D \times D$. Compute D .

$$D = \boxed{5} \quad \checkmark$$

- Assuming a symmetric zero-padding of size $P = 1$, window-size $W = 3$, and stride $S = 3$, compute the top right entry $y_{1,3}$ after applying 2-max pooling to the input matrix $\mathbf{x}$.

$$y_{1,3} = \boxed{2} \quad \checkmark$$

Hint: the solutions are integers.

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Richtig

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Convolutional neural networks

Which statements about convolutional neural networks (CNNs) are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. CNNs usually alternate between convolutional and fully connected layers through their depth.
- ☒ b. ResNet is a famous CNN, whose good learning performance is due to skip connections. ✓
- ☒ c. Applying the same kernel weight matrix to all positions of the input is called weight sharing. ✓
- ☐ d. Each convolutional layer always uses exactly one kernel.
- ☐ e. Pooling decreases the receptive field through the depth of the network.
- ☒ f. With kernel weight matrix $w_{i,j}$, kernel size R , and input image $x_{a,b}$, the feature map pre-activation in CNNs is calculated as $s_{a,b} = \sum_{i=0}^{R-1} \sum_{j=0}^{R-1} w_{i,j} x_{a+i,b+j}$. ✓
- ☒ g. Transfer Learning allows to use pretrained kernels from other networks which were trained on different data sets. ✓
- ☐ h. With input dimension A , kernel size R , padding P , and stride S , the feature map dimension is $(A + R + 2P)/S + 1$.

Frage 9

Teilweise richtig

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Decision trees, random forests, and boosting

Which statements about decision trees, random forests, and boosting are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Information gain is a common splitting criterion for regression.
- ☒ b. A split maximizes the information gain if the produced subsets are completely homogenous. ✓
- ☐ c. Decision tree learning algorithms recursively split the training set into subsets.
- ☒ d. A common measure of variable importance in random forests is the mean accuracy decrease. ✗
- ☒ e. In a random forest, each decision tree uses a random bootstrap subsample of the original data set. ✓
- ☐ f. In boosting, the main idea is that those samples that led to a small error (i.e. were classified well) in an iteration get a larger weight in the next iteration.
- ☐ g. The Extreme regularized Boosting algorithm remains a state-of-the-art supervised learning method on tabular data.
- ☒ h. The maximal entropy of a data set with M classes is $\ln M$. ✓

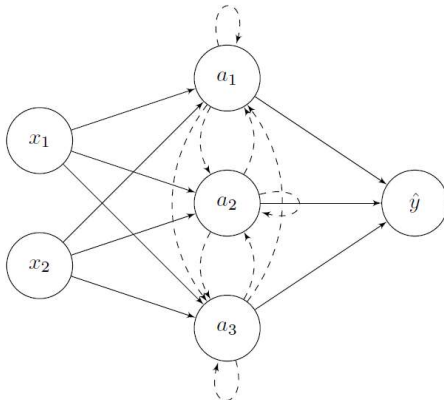
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RNN forward pass

Consider the following recurrent neural network:



The input vector is $\mathbf{x}(t)$. The input weight matrix is denoted as $\mathbf{W}$. The recurrent weight matrix is denoted as $\mathbf{R}$. The output weight matrix is denoted as $\mathbf{V}$. The previous activation in the hidden layer is $\mathbf{a}(t-1)$ and the previous output is $\hat{y}(t-1)$. These are given as:

$$\mathbf{a}(t-1) = (3, 0, 0)^T$$

$$\hat{y}(t-1) = 3$$

$$\mathbf{x}(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\mathbf{W} = \begin{pmatrix} -1 & 3 & 1 \\ 2 & 1 & -1 \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\mathbf{V} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$

The activation function in the hidden layer is Relu. The activation function in the output layer is the sigmoid.

Compute the forward pass producing the pre-activation at the network output $\hat{p}(t)$ and the activation at the network output $\hat{y}(t)$:

• $\hat{p}(t) =$ ✓ (Hint: the answer is integer.)

• $\hat{y}(t) =$ ✓ (Hint: You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Do not give the answer in %. If your Moodle system language is set to English, use dot "." If your Moodle system language is set to German, use comma ",".)

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Logistic regression

Which statements about logistic regression are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Logistic regression has a closed-form solution.
- ☐ b. The likelihood function for a Bernoulli distribution reads: $\mathcal{L}(\{\mathbf{x}, y\}; \mathbf{W}) = \prod_{i=1}^l [g(\mathbf{x}_i; \mathbf{W})^{y_i} + (1 - g(\mathbf{x}_i; \mathbf{W}))^{1-y_i}]$.
- ☐ c. Logistic regression is a non-convex problem as can be seen by the positive-semidefiniteness of the Hesse matrix.
- ☒ d. The momentum term in gradient descent is particularly helpful in flat regions of the loss function. ✓
- ☒ e. In logistic regression, minimization of the loss function can be achieved by gradient descent. ✓
- ☒ f. The softmax function is a generalization of the sigmoid function and is suitable for multi-class classification. ✓
- ☐ g. The logistic function reads: $\sigma(z) = (1 - \exp(z))^{-1}$. It is also known as the sigmoid function.
- ☐ h. Assume that y data are generated around a function $g(x)$ with Gaussian noise. Maximizing the negative log-likelihood of observing the data is equivalent to minimizing the squared error between y and $g(x)$.

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Evaluation measures

Out of 120 total samples, 36 are true negatives and 16 are false negatives. There are twice as many total positives as there are total negatives.

Compute the following evaluation measures:

- Accuracy: ✓
- Balanced-accuracy ✓

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Do not give the answer in %. If your Moodle system language is set to English, use dot "." If your Moodle system language is set to German, use comma ",".

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Constrained optimization

Find the solution w_1^*, w_2^* to the following constrained optimization problem with respect to w_1, w_2 :

$$\begin{aligned} &\text{minimize} && \frac{1}{2}(w_1^2 + 2w_2^2 + 7) \\ &\text{subject to} && w_1 + 6w_2 \geq 19. \end{aligned}$$

- $w_1^* =$ ✓
- $w_2^* =$ ✓

Hint: The solutions are integers.

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Neural networks

Which statements about artificial neural networks are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The vanishing gradient problem can be avoided by the use of suitable loss functions.
- ☐ b. Let $\mathbf{J}^{[l]}$ denote the Jacobian of layer l , where $l = 0$ is the input layer. The recursive formula for delta errors is given by $\delta^{[l+1]T} = \delta^{[l]T} \mathbf{J}^{[l]}$.
- ☒ c. The Universal Approximation Theorem guarantees that neural networks can in principle approximate any continuous function $f \in [0, 1]^m$ arbitrarily well. ✓
- ☒ d. Backpropagation is an instance of reverse mode automatic differentiation. ✓
- ☐ e. Dropout and weight decay are regularization techniques that aim at avoiding underfitting.
- ☐ f. Let us denote the learning rate with $\eta > 0$, the weights with $\mathbf{w}$, and the empirical risk with R . Then the gradient descent update rule reads: $\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} + \eta \nabla_{\mathbf{w}} R$.
- ☐ g. The OR classification problem is a simple classification problem that cannot be solved by a single-layer network.
- ☒ h. Without activation functions, neural networks can only model linear functions. ✓

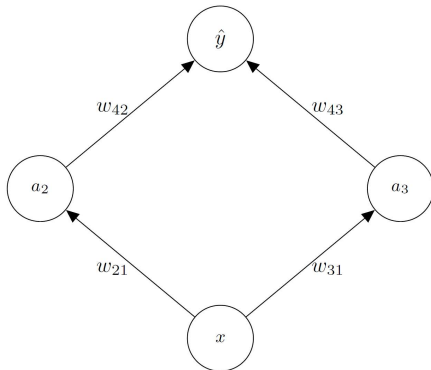
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Richtig

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Backpropagation

Consider the following neural network:



The input unit 1 has activation $a_1 = x$. The preactivations of the hidden units 2 and 3 are denoted as s_2 and s_3 , and their activations are a_2 and a_3 , respectively. The preactivation of the output unit 4 is denoted as s_4 and the output as $a_4 = \hat{y}$. The true label is denoted as y . There are no biases.

In the hidden layer, we use Relu as an activation function, i.e. $f_2(s) = f_3(s) = \text{Relu}(s) = \max(0, s)$. In the output layer, the activation is the identity function.

The delta-error at the output is denoted as δ_4 , and the hidden delta-errors as δ_2 and δ_3 , respectively. Loss function is the squared error, i.e. $L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$.

The learning rate is denoted as η . The network weights are updated via backpropagation and gradient descent, and these updated weights are denoted by the superscript "new".

You are given the input value, the current network weights, the true label, and the learning rate for gradient descent update:

$$x = 2$$

$$w_{21} = 2$$

$$w_{31} = -1$$

$$w_{42} = -1$$

$$w_{43} = 2$$

$$y = 1$$

$$\eta = 0.01$$

Compute the following quantities arising from the forward pass, backpropagation, and gradient descent:

- $\hat{y} =$ ✓
- $\delta_2 =$ ✓
- $\frac{\partial L}{\partial w_{21}} =$ ✓
- $w_{21}^{\text{new}} =$ ✓
- $w_{31}^{\text{new}} =$ ✓

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Do not give the answer in %. If your Moodle system language is set to English, use dot "." If your Moodle system language is set to German, use comma ",".